

ABHINAV VATS

New Delhi, India | +91 9205562868 | abhinavvats.dev@gmail.com
Portfolio | LinkedIn | GitHub | LeetCode

Summary

Computer Science 4th year undergraduate (CGPA: 9.28/10) with experience in full-stack development, AI-powered applications, systems-level programming in Java, REST API design, and cloud deployment on AWS. Skilled in React, Next.js, Node.js, Java, and Python, with hands-on experience building a multithreaded deep-packet-inspection engine, an AI-native CRM platform, and event-driven backend pipelines. First-author IEEE paper under review on CNN/Transformer-based recommendation systems. Seeking a Software Engineer / Full Stack Developer role.

Technical Skills

Languages: Java, JavaScript, TypeScript, Python, SQL, C++

Frontend: React.js, Next.js, Component Architecture, Code Splitting, Lazy Loading

Backend and APIs: Node.js, Express.js, Fastify, FastAPI, Flask, REST APIs, Webhooks, JWT Auth

Systems and Networking: Multithreading, Concurrent Programming, TCP/IP, TLS/SNI, Packet Parsing (Java)

Cloud and DevOps: AWS (EC2, S3, Lambda, ECS), Docker, GitHub Actions CI/CD

Databases: PostgreSQL, Prisma, MongoDB, Redis, Query Optimization, Indexing

Tools: Maven, Postman, Git, GitHub, VS Code

Education

SRM Institute of Science and Technology, Delhi-NCR Campus

B.Tech in Computer Science and Engineering | CGPA: 9.28/10

Expected: May 2027

Aug 2023 – Present

Work Experience

AI Intern

Berry Stenley

Jun 2026 – Jul 2026

- Built and integrated GPT-based automation workflows into internal tools, replacing manual data-handling steps in day-to-day operations.
- Cleaned and structured datasets for downstream model use and supported deployment of AI-assisted features into existing web/app workflows.

UI/UX Frontend Developer Intern

Looping Loons

Nov 2025 – Apr 2026

- Built 50+ reusable React components for an AI-driven platform, cutting feature delivery time by 40%; integrated with Node.js backend services and recommendation workflows.
- Improved page load performance via code splitting and lazy loading, raising Lighthouse scores to 95+; shipped features end-to-end across Agile sprints with product and design teams.

Technical Projects

DPI Engine – Java Deep Packet Inspection System | [GitHub](#)

Java, Maven, pcap4j, Npcap

- Built a Java DPI engine parsing offline PCAP captures and live traffic via pcap4j/Npcap, with Ethernet/IPv4/IPv6/TCP/UDP header parsing and five-tuple flow tracking for stateful connection state.
- Implemented traffic classification via TLS SNI, HTTP Host, DNS, and QUIC extraction, plus configurable IP/app/domain blocking rules with verbose per-packet decision logging.
- Designed a multi-threaded pipeline (load balancers, fast-path workers, thread-safe bounded queues, consistent-hash flow routing) to parallelize packet classification while preserving per-flow ordering.

XenoCopilot – AI-Native CRM Platform | [GitHub](#) | [Live](#)

Next.js, Fastify, PostgreSQL, Prisma, Gemini/Groq APIs

- Built an AI-native CRM translating natural language goals into multi-channel campaigns; engineered a decoupled, event-driven webhook pipeline between a Fastify backend and an isolated delivery microservice, load-tested with simulated network jitter across 3,000+ message events.
- Designed an "Opportunity Engine" on a multi-LLM cascade (Gemini 2.5 Flash to Groq LLaMA 3) with an 8.5s timeout guard and multi-key failover; built an RFM engine grouping users into 5+ cohorts for revenue prediction.

GigShield – Parametric Insurance Platform | [GitHub](#) | [Live](#)

React, Node.js, Express, PostgreSQL, Supabase, Vercel

- Designed a full-stack parametric insurance platform for gig-economy workers; built a real-time risk engine integrating live weather and AQI APIs to drive automated claim triggering without manual claim review.
- Added fraud checks (geolocation, velocity checks, liveness verification) and shipped 15+ React pages, deployed on Vercel with PostgreSQL and Supabase Auth.

Leadership

President, National Service Scheme (NSS) – SRM IST, Delhi-NCR

Jan 2024 – Jun 2025

- Led skill-development workshops and managed 100+ student volunteers across NSS community service initiatives as chapter president.

Certifications and Achievements

- Technology Virtual Experience Program – Deloitte AU (Forage)
- Software Engineering Virtual Experience Program – Accenture Nordics (Forage)